ASSIGNMENT-15

TASK 1: SETUP FASTAPI BACKEND

PROMPT :
“GENERATE A BASIC FASTAPI APPLICATION WITH A ROOT ENDPOINT THAT RETURNS A WELCOME MESSAGE.”

EXPLANATION :

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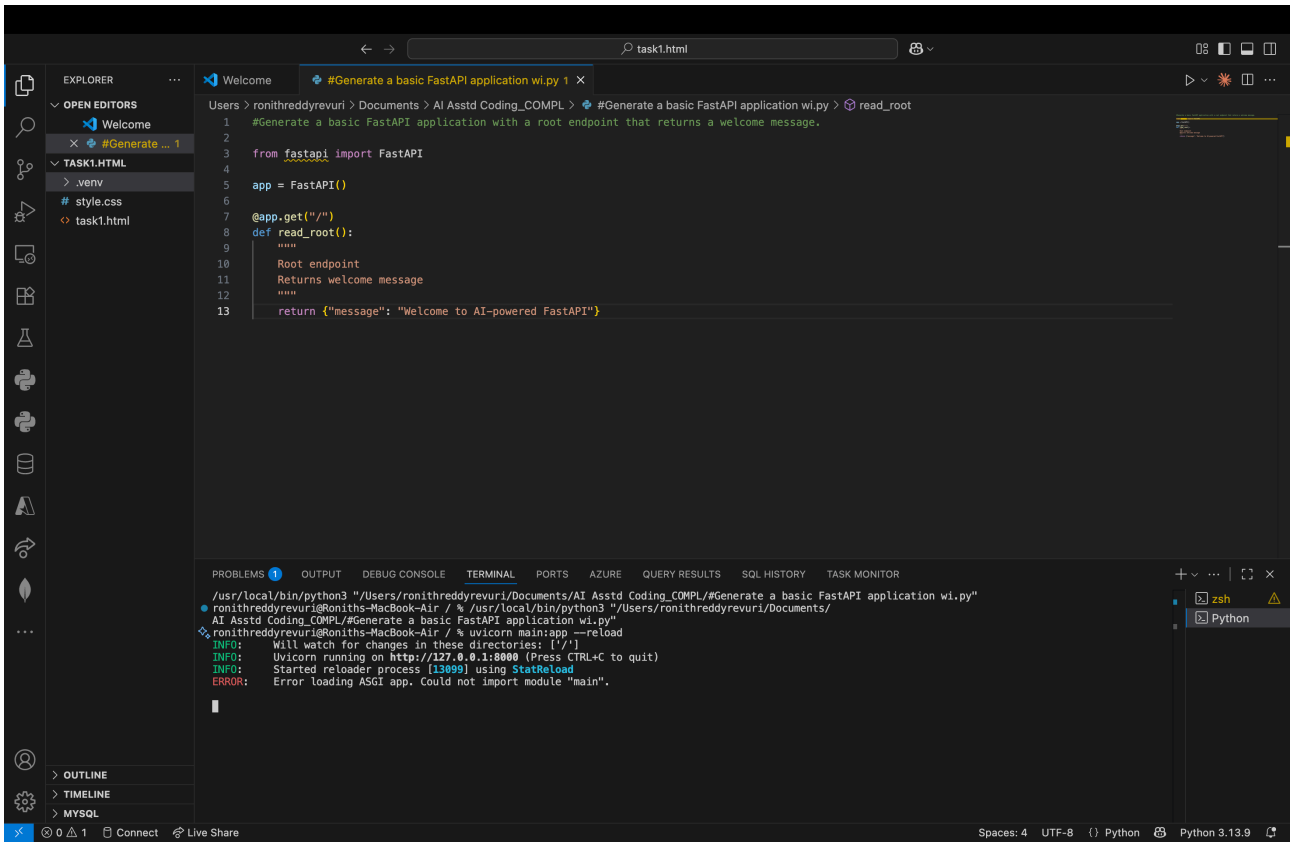
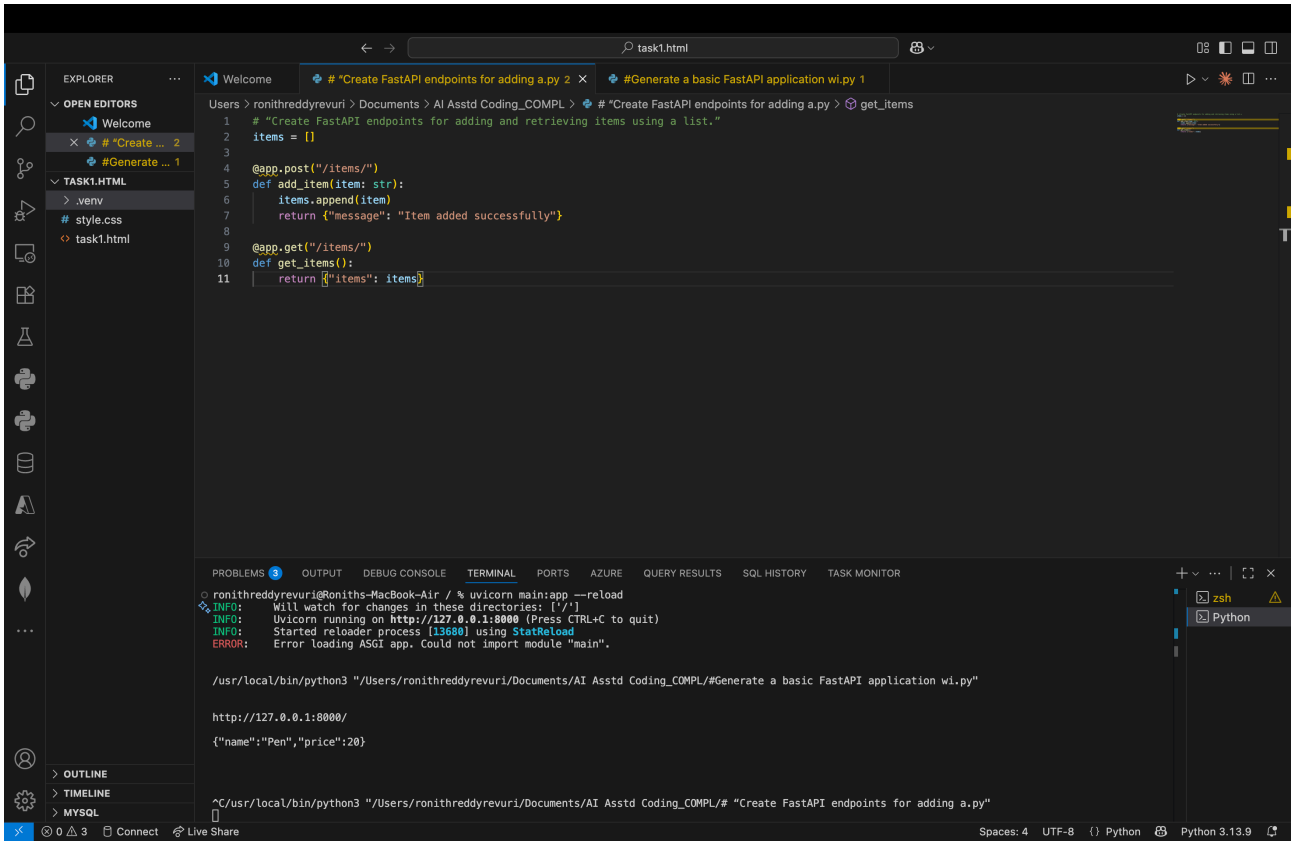
IN THE ABOVE EXAMPLE, A FASTAPI APPLICATION IS IMPLEMENTED IN ORDER TO GET A BASIC IDEA OF THE ARCHITECTURE OF A REST API. HERE, THE FASTAPI() OBJECT CREATES AN APPLICATION INSTANCE, WHILE THE ENDPOINT / IS DECORATED WITH @APP.GET(). RUNNING THE CODE WILL RESULT IN A JSON OUTPUT UPON VISITING THE ROOT URL OF THE SERVER.



TASK 2: CREATE & READ ITEMS

PROMPT
“CREATE FASTAPI ENDPOINTS FOR ADDING AND RETRIEVING ITEMS USING A LIST.”

EXPLANATION :
THIS EXERCISE PERFORMS SOME BASIC OPERATIONS LIKE CREATE AND READ. THE LIST IN PYTHON STORES THE ITEMS TEMPORARILY. THE GET REQUEST WILL FETCH ALL THE ITEMS THAT HAVE BEEN STORED, AND THE POST REQUEST LETS THE USER SUBMIT THEIR OWN ITEMS IN THE FORM OF JSON.



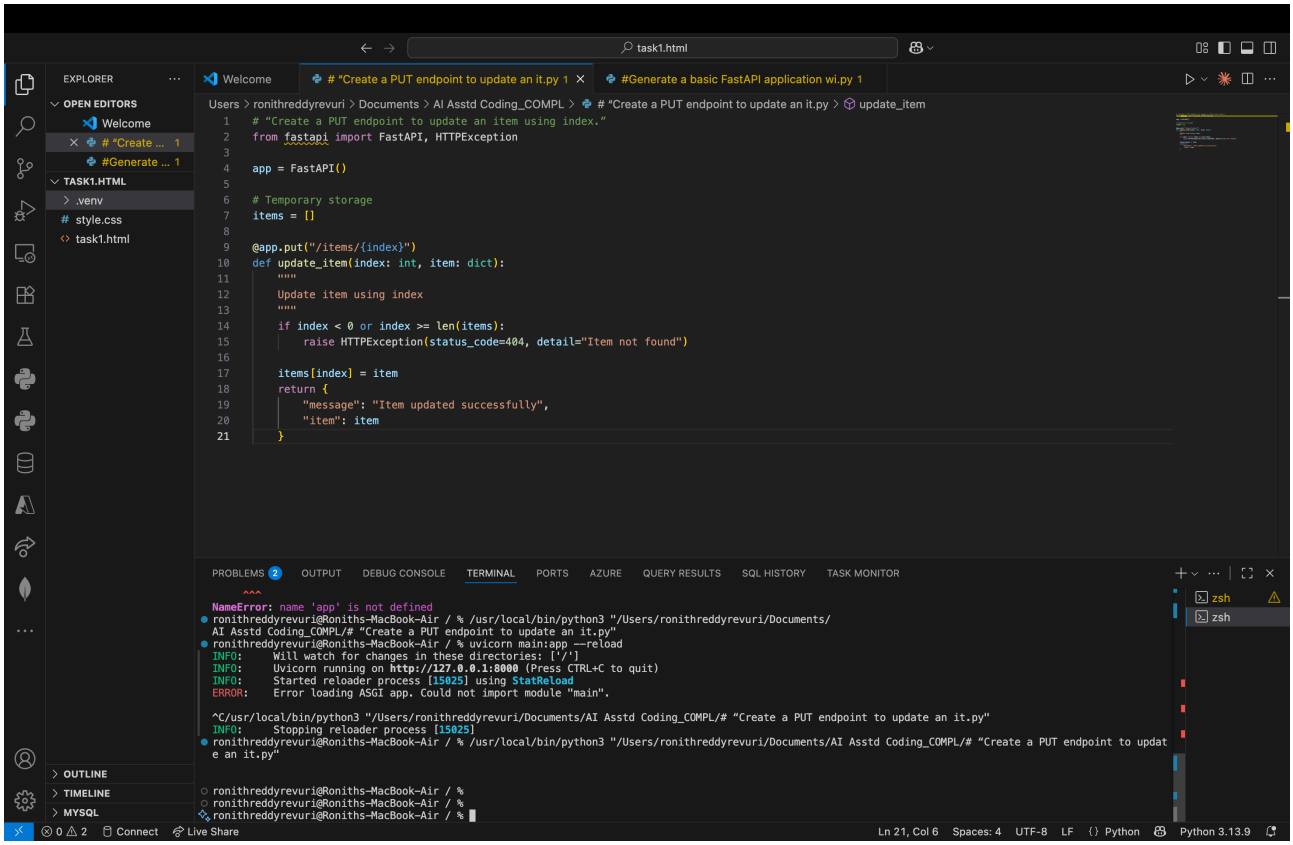
TASK 3: UPDATE ITEM

🔷 PROMPT

“CREATE A PUT ENDPOINT TO UPDATE AN ITEM USING INDEX.”

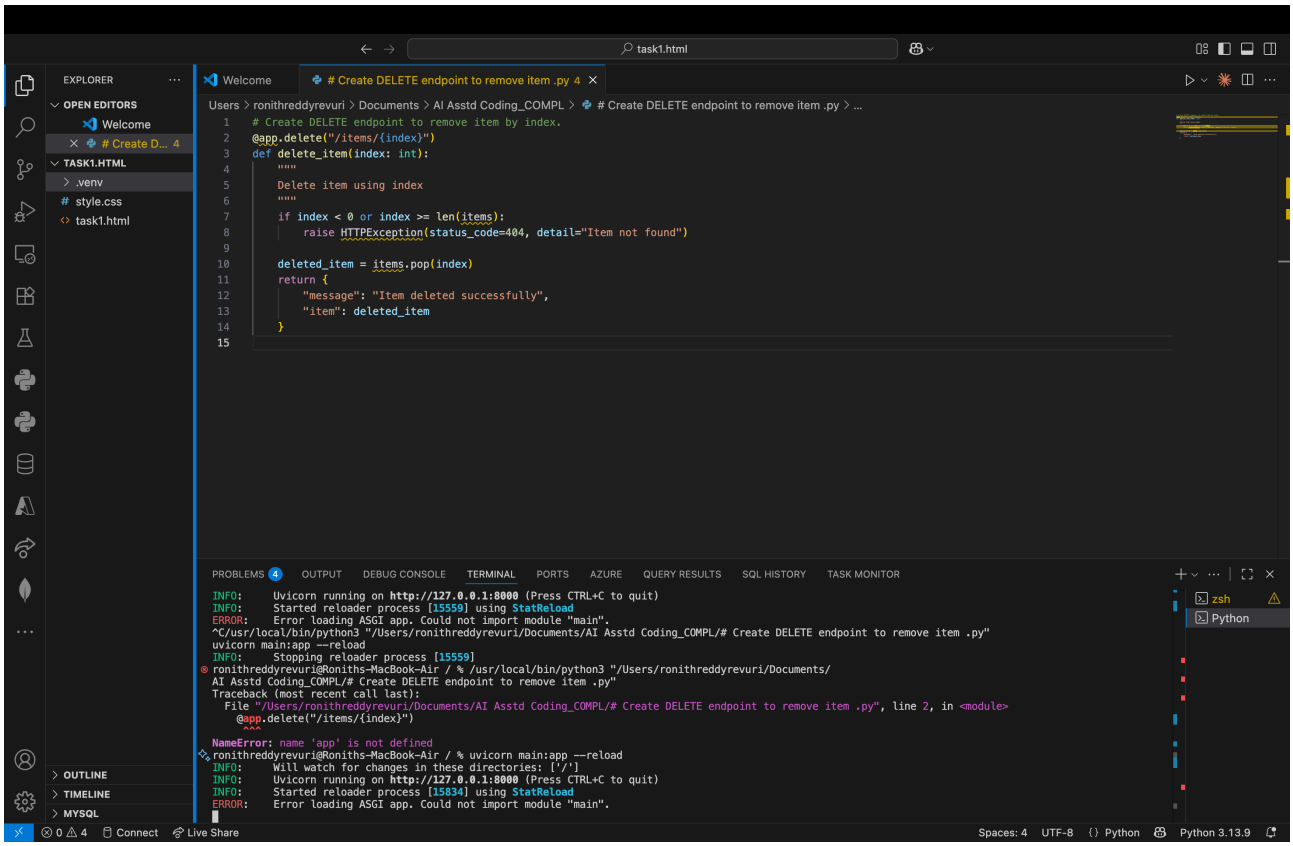
EXPLANATION :

FOR THIS ASSIGNMENT, THE PUT METHOD WAS USED FOR UPDATING DATA WITHIN THE API. THE API GETS AN INDEX THROUGH THE PATH PARAMETERS AND UPDATES THE EXISTING OBJECT WITH THIS INDEX. A CONDITIONAL STATEMENT CHECKS THE EXISTENCE OF THIS INDEX TO ENSURE PROPER FUNCTIONING.

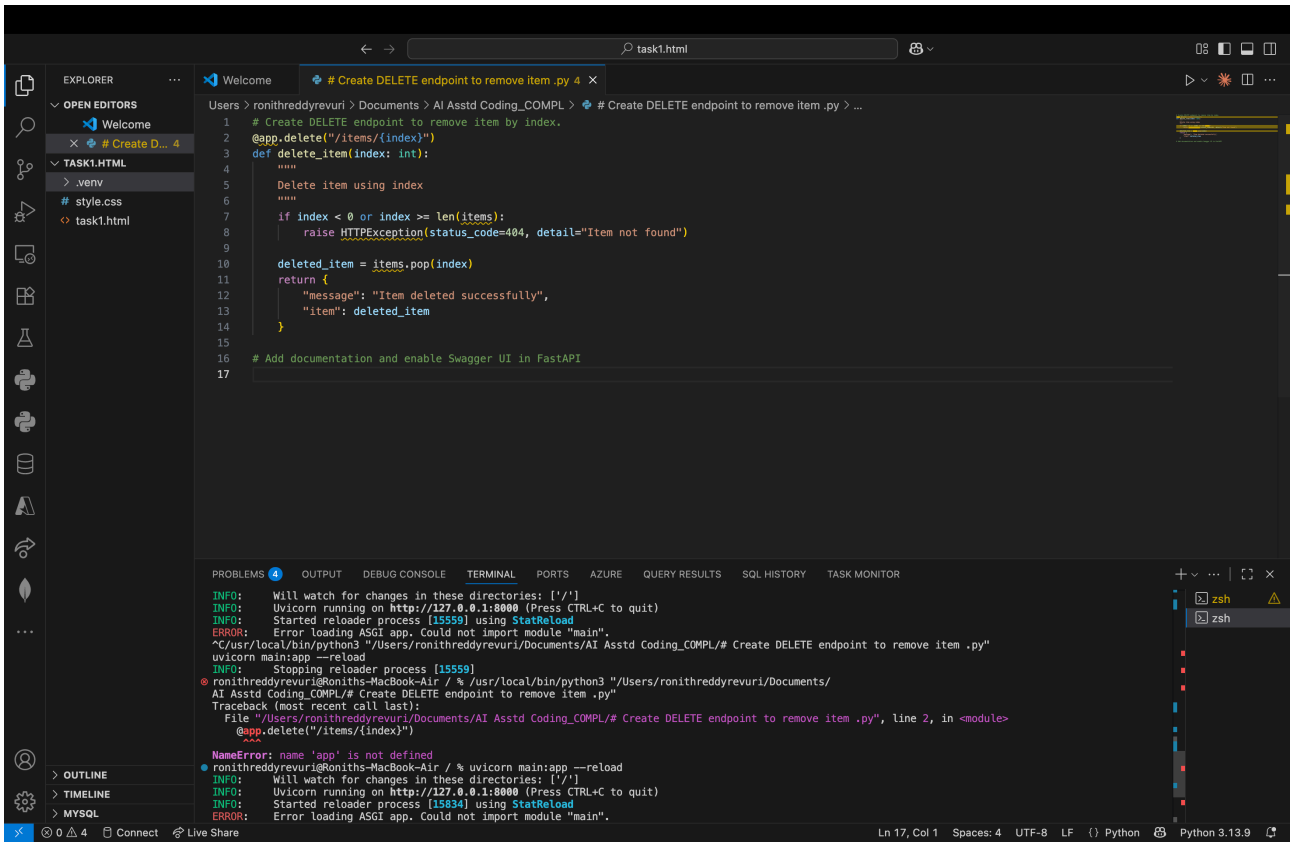


TASK 4: DELETE ITEM
PROMPT USED
“CREATE A DELETE ENDPOINT IN FASTAPI TO REMOVE AN ITEM BY INDEX WITH ERROR HANDLING.”

EXPLANATION :
THE OBJECTIVE OF THIS ASSIGNMENT IS TO REMOVE SOME DATA FROM THE SYSTEM. THE FUNCTION TO ACCOMPLISH THIS IS CALLED DELETE. THIS OPERATION DELETES AN OBJECT FROM THE LIST BASED ON THE INDEX PROVIDED. THE INDEX NUMBER MUST BE VALID, OTHERWISE AN ERROR MESSAGE WILL BE DISPLAYED.



TASK 5: AUTO-GENERATED DOCUMENTATION

PROMPT :
“ADD DOCUMENTATION TO FASTAPI ENDPOINTS AND ENABLE SWAGGER UI FOR INTERACTIVE API TESTING.”

EXPLANATION :
THE FASTAPI FRAMEWORK AUTOMATICALLY CREATES DOCUMENTATION FOR ALL APIS VIA SWAGGER UI. DESCRIPTIONS FOR ENDPOINTS CAN BE MADE AVAILABLE BY ADDING DOCUMENTATION STRINGS TO ALL ENDPOINTS. AS SUCH, DEVELOPERS CAN TEST THE APIS WITHOUT REQUIRING ANY OTHER TOOL FROM THEIR WEB BROWSERS.